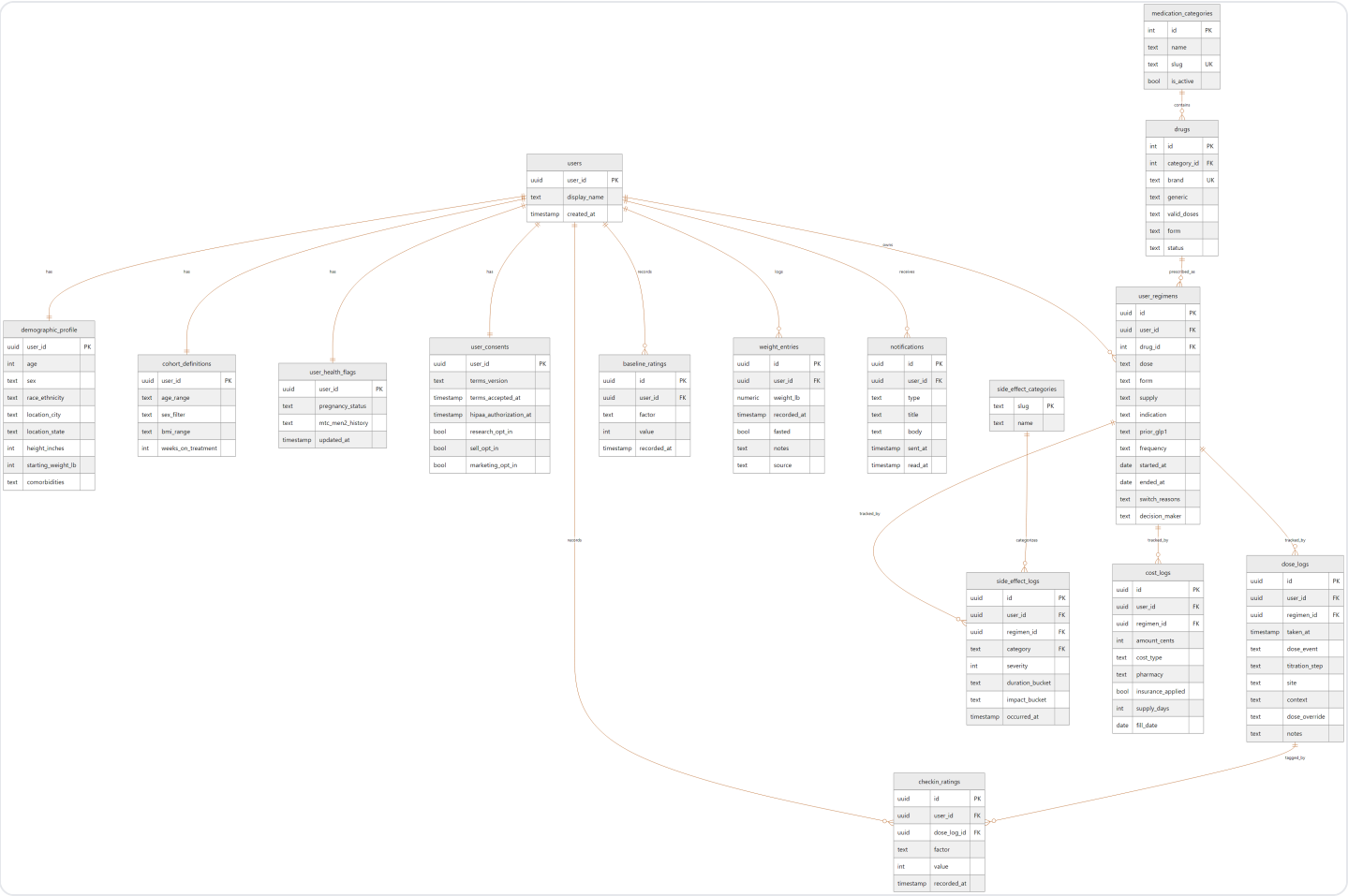
Database schema & per-table documentation

Every table, why it exists, and the fields it holds. Field types omitted — see `schema.sql` for the authoritative Postgres DDL.



ERD rendered from `schema.mmd`. Each box is a table, each line is a FK relationship.

Why each table exists

IDENTITY

users

The one identity row per person. Everything else hangs off of `user_id`.

FIELDS

`user_id`	
`created_at`	
`display_name`	optional greeting-only

WHY

> We need a stable, anonymizable identifier that survives renames and device changes. **No legal first/last name is stored** — only an optional display name for greetings. Email and auth credentials live in a separate identity provider so this table stays HIPAA-aligned under the minimum-necessary principle.

USED BY

`every screen`

IDENTITY · COHORT FOUNDATION

demographic_profile

The fields captured on the Demographics onboarding screen. This is the **single most important table in the app** — every cohort comparison starts here.

FIELDS

`user_id`	1:1 with users
`age`	→ cohort age window
`sex`	F/M/NB/PNTS
`race_ethnicity`	multi-select; optional
`location_city`	
`location_state`	
`height_inches`	→ BMI input
`starting_weight_lb`	→ BMI + weight Δ baseline
`comorbidities`	T2D / PCOS / HTN / CVD / GI / Pancreatitis / Thyroid / None

WHY

> **Age + weight + height** → matched-cohort age and BMI windows (the client's core ask: "how do my results compare to people like me?"). **Sex** → cohort sex filter and GLP-1 response differs by sex. **Race/ethnicity** → optional demographic lens for insights (never used as a treatment gate). **Location** → regional pricing variance on cost comparisons.

USED BY

`/demographics` `/profile` `CohortBadge` `/comparison` `/insights` `/adherence`

COHORT

cohort_definitions

The *editable* cohort filter the user sees on the CohortBadge. Defaults are derived from `demographic_profile`, but the user can tune them from Profile.

FIELDS

user_id	
age_range	default [age−5, age+5]
sex_filter	
bmi_range	default [BMI−2, BMI+2]
category_filter	e.g., [1] for GLP-1s
weeks_on_treatment	

WHY

> Decouples the **cohort lens** from raw demographic facts. A user can broaden their cohort to see more people, or tighten it to get tighter matches — without changing their profile.

USED BY

`/profile (edit)` `CohortBadge` `/comparison` `/insights`

REFERENCE · STATIC

medication_categories

The category tree shown on the Medication onboarding screen (GLP-1s active; Blood pressure / Birth control / Mental health coming soon).

FIELDS

id	
name	"GLP-1s"
slug	"glp-1"
is_active	controls "coming soon" state

WHY

> Adding a new category (e.g., Blood pressure) is a data change — flip `is_active=true`, add its drugs, ship the matching dashboard variant. **No schema change.**

USED BY

`/medication` `/dashboard eyebrow`

REFERENCE · STATIC

drugs

The catalog of medications within each category, with the valid dose strengths for each.

FIELDS

id	
category_id	
brand	"Mounjaro"
generic	"tirzepatide"
valid_doses	["2.5 mg","5 mg",…]

WHY

> Dose options differ per drug (Ozempic doesn't have 15 mg; Mounjaro does). Storing `valid_doses` on the drug row drives the dose dropdown and prevents nonsensical entries.

USED BY

`/medication` `/switch-medication` `/comparison rankings`

USER STATE · HISTORY

user_regimens

One row per regimen the user has ever been on. The row with `ended_at IS NULL` is the current one. Past rows preserve the drug timeline visible on Profile and drive historical side-effect / cost comparisons.

FIELDS

id	
user_id	
drug_id	
dose	e.g., "5 mg"
form	injection / pill
supply	branded / compounded
indication	weight / t2d / both — shapes cohort match
prior_glp1	naive / switched / restarted
frequency	weekly / daily / twice_weekly / other
started_at	
ended_at	null = current
switch_reasons	set on end
decision_maker	self / doctor

WHY

> The medication timeline is a **product feature** (client: "what did I try before, why did I switch?"). By having event logs FK to `regimen_id`, a dose/side-effect/cost event stays tied to the exact drug + dose in use at the time — even after the user switches.

USED BY

`/dashboard regimen card` `/profile history` `/switch-medication` `all log screens`

RATINGS · REFERENCE LINE

baseline_ratings

The 6 one-time ratings captured on the Baselines onboarding screen — the reference line every "Baseline → now" comparison is anchored to.

FIELDS

id	
user_id	
factor	energy / appetite / mood / sleep / activity / digestion
value	
recorded_at	

WHY

> Without a baseline, "You're improving" is meaningless. This table is the anchor for the Dashboard "Baseline → now" grid and Insights' "Your shifts since baseline" card.

USED BY

`/baselines` `/dashboard shifts grid` `/insights shifts card`

RATINGS · ONGOING

checkin_ratings

Ongoing 1–5 ratings the user submits via the Log Dose optional check-in. Same 6 factors as baseline; timestamped per event.

FIELDS

id	
user_id	
dose_log_id	tied to dose if submitted with one
factor	6 baseline + 4 GLP-1 extras (food_tolerance, hydration, muscle_mass, menstrual_changes)
value	
recorded_at	

WHY

> Lets the "Baseline → now" view actually move over time. Tying check-ins to a specific dose helps us correlate subjective shifts with dose timing (e.g., "mood dips 2 days after the weekly Mounjaro injection"). **Supports a wider factor set than baseline** — GLP-1-specific factors (muscle mass, food tolerance, hydration, cycle regularity) are captured here even if the user skipped them at onboarding.

USED BY

`/log-dose (optional check-in)` `/dashboard shifts grid` `/insights shifts card`

EVENT · ADHERENCE

dose_logs

One row per dose taken. This is the source of truth for everything adherence-related.

FIELDS

id	
user_id	
regimen_id	
taken_at	null if skipped
dose_event	on_time / took_late / skipped_side_effects / skipped_other
titration_step	steady / stepped_up / stepped_down
site	injection only: abdomen / thigh / upper_arm
context	pill only: empty_stomach / with_food / after_food
dose_override	if dose changed this time
notes	

WHY

> Adherence % (Dashboard ring, Adherence calendar, Comparison outcome row) is computed from the `dose_event` mix — a skip tied to side effects tells a different story than a simple missed dose. `titration_step` clusters side-effect reports around dose escalations. `site` and `context` are form-conditional, so the same table serves injections and oral GLP-1s.

USED BY

`/log-dose` `/adherence` `/dashboard ring` `/comparison outcomes`

EVENT · TOLERABILITY

side_effect_logs

Per-event capture of a side effect with severity and impact. Feeds the "Side effects by drug" card on Insights and the side-effect-days outcome on Comparison.

FIELDS

id	
user_id	
regimen_id	anchors to drug in use
category	nausea / fatigue / diarrhea / …
severity	
duration_bucket	
impact_bucket	
occurred_at	

WHY

> Aggregated across a cohort, this table answers "how common is nausea on Mounjaro for women age 30–39 in their first 90 days?" — the core data for Insights' "Side effects by drug" card.

USED BY

`/log-side-effect` `/insights side-effects-by-drug` `/insights trends` `/comparison side-effect-days`

EVENT · COST

cost_logs

Per-fill cost record. Feeds personal cost trends and the cohort median used in the Insights "Monthly cost by drug" card.

FIELDS

id	
user_id	
regimen_id	
amount_cents	integer cents, never float
cost_type	copay / oop / coupon / retail
pharmacy	
insurance_applied	
supply_days	normalizes to /month
fill_date	

WHY

> Monthly cost is computed as `amount_cents × (30 / supply_days)` so 90-day fills get normalized. Splitting by `cost_type` lets us surface "with insurance" vs "retail" bands on cohort comparisons.

USED BY

`/log-cost` `/insights cost-by-drug` `/comparison cost row` `/dashboard cost summary`

EVENT · OUTCOME

weight_entries

One row per weigh-in. The time series that powers the Dashboard and Insights weight-trend charts, the "You vs. cohort" weight-change row on Comparison, and the baseline-to-current Δ shown throughout the app.

FIELDS

id	
user_id	
weight_lb	60–700 range check
recorded_at	includes time-of-weigh-in
fasted	captured on Log weight
notes	
source	manual / apple_health / google_fit / smart_scale

WHY

> Weight Δ is the single most asked-about metric in the app — the cohort pitch, the Comparison outcome row, and the Dashboard/Insights trend charts all depend on this time series. Keeping `source` on each row lets us distinguish a manually-entered Monday weigh-in from a Withings auto-sync so we can apply different confidence rules when computing trends. A `unique(user_id, recorded_at)` index prevents duplicate rows from simultaneous writes (e.g., manual entry racing an Apple Health HealthKit push).

USED BY

`/log-weight` `/dashboard weight trend` `/insights weight trend` `/comparison weight change` `clinician PDF export`

SAFETY · SCREENING

user_health_flags

Boxed-warning screening answers captured on the Medication onboarding safety check. Used to surface red-flag warnings and to exclude users from certain cohort-match comparisons where appropriate.

FIELDS

user_id	
pregnancy_status	yes / no / prefer_not_to_say
mtc_men2_history	yes / no / prefer_not_to_say
updated_at	

WHY

> GLP-1s carry a boxed warning for medullary thyroid carcinoma / MEN 2 history and are not recommended during pregnancy. Storing these as structured flags (not free text) lets the app show a **red-flag callout** on Medication onboarding and gate certain cohort comparisons (e.g., a pregnancy-flagged user shouldn't see a "people like you lost X%" claim that doesn't apply). Separate from `demographic_profile` so we can RLS-protect these columns more tightly.

USED BY

`/medication safety check` `/comparison cohort filter` `/profile privacy view`

CONSENT · PRIVACY

user_consents

Persistent record of the user's legal agreements and data-sharing preferences, captured at onboarding and editable in Profile.

FIELDS

user_id	
terms_version	e.g., "2026-04-01"
terms_accepted_at	
privacy_version	
privacy_accepted_at	
hipaa_authorization_at	
research_opt_in	default TRUE
sell_opt_in	default FALSE · opt-in required
marketing_opt_in	default FALSE
updated_at	

WHY

> Regulatory audit trail — we need to prove exactly which version of Terms/Privacy/HIPAA a given user agreed to, and when. Opt-in booleans gate data handling downstream: `sell_opt_in=FALSE` means the user's rows are excluded from any aggregate dataset sold or shared with commercial third parties (CCPA / CPRA / Washington MHMDA compliance). Versioned text columns let us re-prompt users when terms change without losing the historical record.

USED BY

`/consent` `/profile privacy view` `downstream data-sharing jobs`

OUTPUT

notifications

Outbound messages shown on the Notifications screen and as dashboard bell badges.

FIELDS

id	
user_id	
type	reminder / cohort_insight / weekly_summary / confirmation
title	
body	
sent_at	
read_at	null = unread dot

WHY

> Separating the message from the data generating it (e.g., a cohort insight is derived from other tables) lets the notification feed be rebuilt without losing history of what the user actually saw.

USED BY

`/notifications` `/dashboard bell badge`

WORKED EXAMPLE

How a cohort query uses these tables

To compute the "people like you" figures on Comparison and Insights, the backend joins a user's `cohort_definitions` row against every other user's `demographic_profile`, then aggregates that peer set's `dose_logs`, `side_effect_logs`, and `cost_logs` for the current category.

```
-- For user U, find matched peers + their median 12-week weight change on Mounjaro
WITH my_cohort AS (
  SELECT dp.user_id
  FROM demographic_profile dp
  JOIN cohort_definitions cd ON cd.user_id = :me
  WHERE dp.age BETWEEN lower(cd.age_range) AND upper(cd.age_range)
    AND dp.sex = ANY(cd.sex_filter)
    AND bmi(dp.height_inches, dp.starting_weight_lb) <@ cd.bmi_range
)
SELECT percentile_cont(0.5) WITHIN GROUP (ORDER BY weight_change_pct)
FROM weight_change_12wk_mv
WHERE user_id IN (SELECT user_id FROM my_cohort)
  AND drug_id = :mounjaro_id;
```

In practice, heavy cohort aggregates (12-week weight change, side-effect incidence by drug, cost medians) live in materialized views refreshed nightly so the app itself only reads cheap summaries. The tables above are the raw sources of truth those views compute from.

DATA DICTIONARY

Every variable — type, use, sensitivity

Cross-reference for every field. **Sensitivity** maps to HIPAA PHI where applicable; PII covers locational / financial data.

PHI PII / financial Non-sensitive

public.users

Identity wrapper. FK to auth.users. 1 row per person.

Variable	Type	Use	Sensitivity
user_id	uuid PK	Auth linkage; FK target for every user-owned row.	PHI identifier
created_at	timestamptz	Audit / cohort longevity.	Non-sensitive
updated_at	timestamptz	Cache invalidation, optimistic concurrency.	Non-sensitive
display_name	text	UI greeting ("Good morning, Alex").	PII

public.demographic_profile

Cohort foundation. Set once on /demographics onboarding.

Variable	Type	Use	Sensitivity
user_id	uuid PK/FK	1:1 with users.	PHI identifier
age	smallint	Cohort matching window (±5y default).	PHI
sex	enum	Cohort filter; GLP-1 response differs by sex.	PHI · protected class
race_ethnicity	text[]	Optional cohort lens on Insights. Never a treatment gate.	PHI · protected class
location_city	text	Regional pricing variance on cost comparisons.	PII
location_state	char(2)	State-level cohort filter, pricing.	PII
height_inches	smallint	BMI input → cohort BMI window.	PHI
starting_weight_lb	smallint	BMI input; baseline for weight Δ trend.	PHI
created_at	timestamptz	Audit.	Non-sensitive
updated_at	timestamptz	Cache invalidation.	Non-sensitive

public.cohort_definitions

Editable cohort lens. Auto-seeded from demographic_profile.

Variable	Type	Use	Sensitivity
user_id	uuid PK/FK	1:1 with users.	PHI identifier
age_min · age_max	smallint	Cohort age window (default age ±5).	PHI-adjacent
sex_filter	enum[]	Which sex(es) to include in cohort.	Derived filter
bmi_min · bmi_max	numeric(4,1)	Cohort BMI window (default BMI ±2).	PHI-adjacent
category_filter	int[]	Which medication_categories to include.	Non-sensitive
weeks_on_treatment	smallint	Matched-cohort treatment window (default 12).	Non-sensitive

public.medication_categories

Static reference catalog. World-readable.

Variable	Type	Use	Sensitivity
id	int PK	Category lookup.	Non-sensitive
name	text	"GLP-1s" etc — UI display.	Non-sensitive
slug	text	Stable URL key.	Non-sensitive
is_active	bool	Gates "Coming soon" vs active category cards.	Non-sensitive

public.drugs

Static reference catalog. World-readable.

Variable	Type	Use	Sensitivity
id	int PK	Drug lookup.	Non-sensitive

Variable	Type	Use	Sensitivity
category_id	int FK	Taxonomy link.	Non-sensitive
brand	text	"Mounjaro", "Ozempic".	Non-sensitive
generic	text	"tirzepatide", "semaglutide".	Non-sensitive
valid_doses	text[]	Drives dose dropdown on /medication + /switch.	Non-sensitive

public.user_regimens

Current + historical regimens. One active row per user (ended_at IS NULL).

Variable	Type	Use	Sensitivity
id	uuid PK	Regimen identity; FK target for event logs.	PHI
user_id	uuid FK	Ownership.	PHI
drug_id	int FK	Which drug.	PHI
dose	text	Strength, e.g., "5 mg".	PHI
frequency	enum	Weekly / daily / etc.	PHI
started_at	date	Regimen start for duration calc.	PHI
ended_at	date?	Null = active. Set on switch.	PHI
switch_reasons	text[]?	Why the user switched. Cohort insight input.	PHI
decision_maker	enum?	Self vs doctor.	PHI
created_at · updated_at	timestamptz	Audit, cache.	Non-sensitive

public.baseline_ratings

6 one-time ratings set on /baselines onboarding.

Variable	Type	Use	Sensitivity
id	uuid PK	Row identity.	PHI
user_id	uuid FK	Ownership.	PHI
factor	enum	energy / appetite / mood / sleep / activity / digestion.	Non-sensitive
value	int 1–5	Baseline rating. Anchor for all shift deltas.	PHI · PRO
recorded_at	timestamptz	When baseline was captured.	PHI-adjacent

public.checkin_ratings

Ongoing 1–5 ratings from Log Dose optional check-in.

Variable	Type	Use	Sensitivity
id	uuid PK	Row identity.	PHI
user_id	uuid FK	Ownership.	PHI
dose_log_id	uuid FK?	Correlates check-in to a specific dose.	PHI
factor	enum	Which baseline factor.	Non-sensitive
value	int 1–5	Current rating. Drives "Baseline → now" grid.	PHI · PRO
recorded_at	timestamptz	When rating was taken.	PHI-adjacent

public.dose_logs

One row per dose taken. Source of truth for adherence.

Variable	Type	Use	Sensitivity
id	uuid PK	Event identity.	PHI
user_id · regimen_id	uuid FK	Composite FK ties event to correct regimen.	PHI
taken_at	timestamptz	Adherence % computation.	PHI
site	enum	Injection site rotation for clinician reports.	PHI

Variable	Type	Use	Sensitivity
dose_override	text?	If dose changed this time.	PHI
notes	text?	Free-text observations. **Candidate for PreVeil at v2** — clinical free text has the highest re-identification risk.	PHI · free text
created_at	timestamptz	Audit (when logged, vs taken_at = when event happened).	Non-sensitive

public.side_effect_logs

Per-event tolerability capture. Feeds Insights "Side effects by drug".

Variable	Type	Use	Sensitivity
id	uuid PK	Event identity.	PHI
user_id · regimen_id	uuid FK	Composite FK anchors side effect to drug in use.	PHI
category	enum	nausea / fatigue / diarrhea / …	PHI · medical
severity	int 1–5	Severity scale.	PHI
duration_bucket	enum	<1h / 1–4h / 4–12h / 12–24h / ongoing.	PHI
impact_bucket	enum	Daily-activity impact.	PHI
occurred_at	timestamptz	Correlates to dose timing.	PHI
created_at	timestamptz	Audit.	Non-sensitive

public.cost_logs

Per-fill cost. Normalizes to /month; powers Insights "Monthly cost by drug".

Variable	Type	Use	Sensitivity
id	uuid PK	Event identity.	PHI
user_id · regimen_id	uuid FK	Composite FK.	PHI
amount_cents	int	What the user paid (integer cents).	Financial
cost_type	enum	copay / out_of_pocket / coupon / retail.	Financial
pharmacy	text?	Optional. Where filled.	PII
insurance_applied	bool	Split with-insurance vs retail bands.	Financial
supply_days	smallint	30 / 60 / 90 — normalizes to monthly.	Non-sensitive
fill_date	date	When the fill happened.	PHI-adjacent
created_at	timestamptz	Audit.	Non-sensitive

public.notifications

Outbound messages on /notifications. Backend-written only.

Variable	Type	Use	Sensitivity
id	uuid PK	Message identity.	Non-sensitive
user_id	uuid FK	Who receives it.	PHI
type	enum	reminder / cohort_insight / weekly_summary / confirmation.	Non-sensitive
title · body	text	May reference cohort %, adherence, drug names.	PHI (possible)
sent_at · read_at	timestamptz	Unread badge, engagement telemetry.	Non-sensitive

Derived outputs · not schema rows

Assembled on-demand from the tables above. These are the PreVeil touchpoints.

Output	Format	Storage / transport	Use	Sensitivity
Clinician PDF report	PDF	Summary of regimen, adherence, side-effect timeline, cost for provider review. Triggered from Insights → Export → PDF Report.	PHI · full dossier	

Output	Format	Storage / transport	Use	Sensitivity
CSV data export	CSV	Raw event logs for clinician / researcher. Triggered from Insights → Export → CSV.	PHI · full dossier	
Cohort aggregates (mat. views)	numeric	Nightly-refreshed materialized views (weight Δ median, side-effect incidence, cost median by drug). No user_id — safe for public read.	Aggregate only	
Cohort insights shown in-app	numeric	Numbers on Comparison / Insights (e.g., "n=412 matched users"). Read directly from the materialized views.	Aggregate only	
Clinician messaging (future)	thread	If we add patient↔provider messaging, route through PreVeil — not Supabase.	PHI · clinical detail	

Rules of thumb

Question	Rule
Does this field belong in Supabase or PreVeil?	Structured data (enums, numerics, timestamps, FKs) → **Supabase** with RLS. Free-text clinical notes or whole-dossier exports → **PreVeil**.
Is this PHI?	If it's a 1:1 fact about the patient's body or treatment (age, weight, drug, dose, adherence, side effect), yes. If it's a shared catalog (drug brand, category), no.
Should this be shown cross-user?	Only through aggregate materialized views that contain *no* `user_id`. RLS + `SECURITY DEFINER` functions enforce this.
What about deletions?	ON DELETE CASCADE from `users` removes every dependent row. PreVeil-stored exports need a separate deletion policy — PreVeil's admin console handles that.